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Worksheet-7

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Subject Name: Technical Training
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Aim of the Session:

The aim of this practical session is to apply the concept of Stored Procedures in database operations to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

Objective of the Session:

The objectives of this practical session are:

- To understand the concept of stored procedures in DBMS.
- To perform CRUD (Create, Read, Update, Delete) operations using procedures.
- To improve performance using precompiled SQL statements.
- To enhance data security and controlled access.
- To promote code reusability and modular programming.

Practical / Experiment Steps:

Step 1: Creating a table for demonstration.

- Implementing a table to store employee records.

Step 2: Creating a Stored Procedure for Insertion.

- Creating a procedure to insert new employee records.

Step 3: Creating a Stored Procedure for Update.

- Creating a procedure to update employee salary.

Step 4: Creating a Stored Procedure for Deletion.

- Creating a procedure to delete employee records.

Step 5: Creating a Stored Procedure for Retrieval.

- Creating a procedure to fetch employee data.

Procedure of the Practical:

1. Start the system and log in to the computer.
2. Open the PostgreSQL client tool (psql / pgAdmin).
3. Create Tables.

```
CREATE TABLE Employees (
   EmpID SERIAL PRIMARY KEY,
   EmpName VARCHAR(50),
   Salary DECIMAL(10,2),
   Dept VARCHAR(50)
);
```

4. Insert Sample Data

```
INSERT INTO Employees (EmpName, Salary, Dept) VALUES
('Amit', 50000, 'IT'),
('Riya', 45000, 'HR');
```

5. Stored Procedure for INSERT

```
CREATE OR REPLACE PROCEDURE AddEmployee(
   p_name VARCHAR,
   p_salary DECIMAL,
   p_dept VARCHAR
)
LANGUAGE plpgsql
AS $$
BEGIN
   INSERT INTO Employees (EmpName, Salary, Dept)
   VALUES (p_name, p_salary, p_dept);
END;
$$;

CALL AddEmployee('Kunal', 60000, 'Finance');
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	empid [PK] integer	empname character varying (50)	salary numeric (10,2)	dept character varying (50)
1	1	Amit	50000.00	IT
2	2	Riya	45000.00	HR
3	3	Kunal	60000.00	Finance

6. Stored Procedure for UPDATE

```
CREATE OR REPLACE PROCEDURE UpdateSalary(
    p_id INT,
    p_salary DECIMAL
)
LANGUAGE plpgsql
AS $$
BEGIN
    UPDATE Employees
    SET Salary = p_salary
    WHERE EmpID = p_id;
END;
$$;

CALL UpdateSalary(1, 55000);
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	empid [PK] integer	empname character varying (50)	salary numeric (10,2)	dept character varying (50)
1	2	Riya	45000.00	HR
2	3	Kunal	60000.00	Finance
3	1	Amit	55000.00	IT

7. Stored Procedure for DELETE

```
CREATE OR REPLACE PROCEDURE DeleteEmployee(
    p_id INT
)
LANGUAGE plpgsql
AS $$
BEGIN
    DELETE FROM Employees
    WHERE EmpID = p_id;
END;
$$;

CALL DeleteEmployee(2);
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	empid [PK] integer	empname character varying (50)	salary numeric (10,2)	dept character varying (50)
1	3	Kunal	60000.00	Finance
2	1	Amit	55000.00	IT

8. Verify the output after execution.
9. Note down the results obtained.
10. Save the work and take screenshots for record.

I/O Analysis:

Input Provided:

- SQL stored procedure definitions
- Employee table data
- Procedure calls for CRUD operations

Output Generated:

- New records inserted into table

- Updated employee salary
- Deleted records from table
- Retrieved employee data

Learning Outcomes:

- Understanding the concept of stored procedures
- Ability to perform CRUD operations using procedures
- Improved database performance and security
- Code reusability and modular approach
- Practical exposure to real-world database applications.